

Matvei Borodin

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Summary

ML Engineer with a software engineering background, focused on LLMs, reinforcement learning, and efficient inference. Interested in turning research ideas into reliable systems, with an emphasis on rigorous evaluation and balancing model quality with production constraints.

Experience

Alice Voice Activation Team

Moscow, Russia

ML Engineer Intern, Yandex

May 2026 — Present

- Developed a compact **hybrid CNN–Transformer architecture** for on-device **keyword spotting** and implemented **FlashAttention** with **QKNorm**, **relative position biases**, and **RoPE**.
- Built an end-to-end converter from **FP32 PyTorch** to fully quantized **TFLite INT8** without quality degradation, and implemented **KV-cached streaming inference** for on-device deployment.
- Achieved a **15% relative improvement** in target **quality metrics**, reduced compute, and kept **memory usage and inference latency** within production budgets.
- Developed an **embedded C++ orchestration layer** to centrally manage **on-device models** and coordinate their activation.

Alignment Team for Alice Search Generative Answers

Moscow, Russia

ML Engineer Intern, Yandex

December 2025 — March 2026

- Implemented and launched a **GRPO training pipeline** for the RAG agent powering Alice Search generative answers, including runtime infrastructure for **online reward computation** and integration with the alignment pipeline.
- Built core alignment and evaluation components: integrated a new **fact-checking reward**, enabled quantized reward models on supported GPUs, and corrected the aggregation of search-planner contexts used by the reward pipeline.
- Ran and analyzed experiments with **CERL**, **DPO**, and **GRPO**, studying the effects of training-data composition and hyperparameters; diagnosed **reward hacking** and quality regressions through side-by-side assessor evaluations.

MultiHopBench: A Constraint-Aware Benchmark for Factual QA

Moscow, Russia

YandexGPT Pretraining Analytics, YSDA (School of Data Analysis)

October 2025 — June 2026

- Designed a bilingual Russian–English benchmark with **1,300 factual QA examples** for multi-hop retrieval, simultaneous constraint satisfaction, list completion, the **Reversal Curse**, and zero-gold no answer behavior.
- Built a reproducible **Wikidata/SPARQL pipeline** and a two-stage evaluation system in which a schema constrained **LLM judge** produces atomic annotations and a **deterministic scorer** computes auditable coverage, constraint, hallucination, and refusal metrics.

Education

ITMO University, AI Talent Hub

Saint Petersburg, Russia

Master's degree in Artificial Intelligence

2026 — Present

Higher School of Economics

Moscow, Russia

Bachelor's degree, Faculty of Computer Science

2022 — 2026

Technical Skills

Languages: Python, C++, Swift, Kotlin, Java

Technologies: PyTorch, SQL, Nirvana (yandex apache airflow analog), model quantization

Skills: Deep Learning, NLP, LLMs, RL (CERL, DPO, GRPO), RAG, Speech